

NEET Biology Chapter 1 PYQs

Premium Practice Module (30 Questions)

Q1. Which statement correctly defines reproduction as per NCERT?

- A. Biological process by which organisms produce offspring
- B. Increase in body size of an organism
- C. Formation of new tissues in body
- D. Progression through life stages

Q2. Reproduction is considered essential primarily because it ensures continuity of

- A. Individual lifespan
- B. Species generation after generation
- C. Metabolic processes in cells
- D. Growth of multicellular body

Q3. Which of the following events is fundamentally associated with reproduction at cellular level?

- A. Increase in cytoplasmic volume
- B. Differentiation of somatic cells
- C. Replication and transmission of DNA
- D. Accumulation of reserve food

Q4. Assertion A Reproduction is not essential for survival of an individual organism Reason R An organism can complete its lifespan without reproducing

- A. Both A and R are true and R is the correct explanation of A
- B. Both A and R are true but R is not the correct explanation of A
- C. A is true and R is false
- D. A is false and R is true

Q5. Which example best shows that reproduction and growth are not synonymous?

- A. Elongation of root in seedling
- B. Increase in leaf number in plant
- C. Increase in body mass of animal
- D. Formation of new individual in Hydra

Q6. Worker bees in a hive are reproductively

- A. Sterile individuals
- B. Fertile females
- C. Parthenogenetic forms
- D. Sexual males

Q7. Mules are evolutionarily insignificant in terms of heredity because they

- A. Undergo asexual reproduction
- B. Do not contribute genes to next generation
- C. Have shorter lifespan
- D. Show rapid growth

Q8. Which statement correctly distinguishes reproduction from metabolism?

- A. Reproduction increases energy release
- B. Metabolism ensures gene continuity
- C. Reproduction results in offspring formation whereas metabolism sustains life
- D. Metabolism produces gametes

Q9. Assertion A In unicellular organisms cell division is equivalent to reproduction Reason R Cell division in unicellular organisms increases the number of individuals

- A. Both A and R are true and R is the correct explanation of A
- B. Both A and R are true but R is not the correct explanation of A
- C. A is true and R is false
- D. A is false and R is true

Q10. Which of the following organisms is alive but does not reproduce?

- A. Amoeba
- B. Paramecium
- C. Yeast
- D. Mule

Q11. Biologically reproduction is linked with heredity because it involves

- A. Transmission of genetic material from parent to offspring
- B. Increase in cytoplasmic mass
- C. Formation of identical tissues
- D. Growth of somatic cells

Q12. Which statement correctly reflects species level importance of reproduction?

- A. It enhances metabolic rate
- B. It prevents extinction of species
- C. It increases body complexity
- D. It delays senescence

Q13. Assertion A Reproduction always leads to genetic variation Reason R Genetic variation arises mainly during sexual reproduction

- A. Both A and R are true and R is the correct explanation of A
- B. Both A and R are true but R is not the correct explanation of A
- C. A is true and R is false

D. A is false and R is true

Q14. Which of the following is common to both sexual and asexual reproduction?

- A. Involvement of two parents
- B. Gamete formation
- C. Production of offspring
- D. Meiotic division

Q15. Population increase in bacteria culture primarily indicates

- A. Growth of one individual cell
- B. Extension of lifespan
- C. Increase in cell size
- D. Repeated reproduction through cell division

Q16. Assertion A Some individuals in a species may be sterile Reason R Species continuity is maintained by other reproducing members

- A. Both A and R are true and R is the correct explanation of A
- B. Both A and R are true but R is not the correct explanation of A
- C. A is true and R is false
- D. A is false and R is true

Q17. Which situation best demonstrates that reproduction is a species level process?

- A. A single plant increasing height
- B. An organism dying after long life
- C. A population persisting across generations
- D. A cell enlarging in size

Q18. From evolutionary perspective the ultimate significance of reproduction is

- A. Increase in biomass
- B. Maintenance of metabolism
- C. Transmission of genes to next generation
- D. Delay in ageing

Q19. Which event is indispensable for biological reproduction in all organisms?

- A. Accurate duplication of genetic material
- B. Increase in protein synthesis
- C. Respiratory activity
- D. Storage of nutrients

Q20. Why is reproduction not considered synonymous with growth in multicellular organisms?

- A. Growth involves meiosis
- B. Reproduction occurs in all somatic cells

- C. Growth produces genetic variation
- D. Growth increases size whereas reproduction forms a new individual

Q21. Which phase of lifespan is absent in organisms that die immediately after reproduction?

- A. Post reproductive phase
- B. Pre reproductive phase
- C. Juvenile phase
- D. Growth phase

Q22. Which organism is correctly matched with its approximate lifespan as mentioned in NCERT?

- A. Crow – 140 years
- B. Parrot – approximately 140 years
- C. Butterfly – 60 years
- D. Elephant – several thousand years

Q23. Which organism is known to have a lifespan of only a few weeks as per NCERT examples?

- A. Crow
- B. Elephant
- C. Fruit fly
- D. Banyan tree

Q24. Assertion A Lifespan of organisms varies from species to species Reason R Lifespan is independent of environmental conditions

- A. Both A and R are true and R is the correct explanation of A
- B. Both A and R are true but R is not the correct explanation of A
- C. A is true and R is false
- D. A is false and R is true

Q25. Which organism is reported in NCERT to live for several thousand years?

- A. Elephant
- B. Crow
- C. Parrot
- D. Banyan tree

Q26. The upper limit of lifespan in a species is primarily determined by

- A. Genetic constitution
- B. Climatic conditions
- C. Food supply alone
- D. Predation rate alone

Q27. Lifespan of an organism includes which of the following phases?

- A. Only reproductive phase
- B. Pre reproductive reproductive and post reproductive phases
- C. Only juvenile and reproductive phases
- D. Only senescent phase

Q28. Which organism has an approximate lifespan of about 15 years as per NCERT?

- A. Butterfly
- B. Crow
- C. Parrot
- D. Elephant

**Q29. Assertion A Individuals of the same species generally show similar lifespan range
Reason R Lifespan is a characteristic feature of a species**

- A. Both A and R are true and R is the correct explanation of A
- B. Both A and R are true but R is not the correct explanation of A
- C. A is true and R is false
- D. A is false and R is true

Q30. Which organism has an approximate lifespan of 60 to 70 years according to NCERT?

- A. Crow
- B. Parrot
- C. Butterfly
- D. Elephant

Answer Key

Q1: A
Q2: B
Q3: C
Q4: A
Q5: D
Q6: A
Q7: B
Q8: C
Q9: A
Q10: D
Q11: A
Q12: B
Q13: D
Q14: C
Q15: D
Q16: A
Q17: B
Q18: C
Q19: A
Q20: D
Q21: A
Q22: B
Q23: C
Q24: C
Q25: D
Q26: A
Q27: B
Q28: B
Q29: A
Q30: D

Detailed Solutions

Q1: A

NCERT defines reproduction as the biological process by which organisms produce offspring similar to themselves.

Q2: B

NCERT clearly states reproduction is essential for continuity of species and not for survival of individual organisms.

Q3: C

DNA replication and its accurate transmission to offspring is the basic cellular event underlying reproduction.

Q4: A

Individual survival is independent of reproduction and many organisms live full lifespan without reproducing hence reproduction is a species level necessity.

Q5: D

Growth increases size or mass whereas reproduction results in formation of a new individual.

Q6: A

NCERT mentions sterile worker bees showing that some individuals in a species may not reproduce.

Q7: B

Mules are sterile hybrids and cannot transmit genes further hence do not contribute genetically.

Q8: C

Metabolism maintains life processes but reproduction produces new individuals ensuring continuity.

Q9: A

In unicellular organisms each cell division produces a new individual thereby increasing population size.

Q10: D

Mule is a sterile hybrid and does not reproduce though it is alive.

Q11: A

Heredity requires transfer of genetic material which occurs during reproduction.

Q12: B

Without reproduction species would become extinct hence it is vital at species level.

Q13: D

R is true because meiosis and fertilisation in sexual reproduction introduce variation

Q14: C

Both modes ultimately produce new individuals though mechanisms differ.

Q15: D

Bacterial population rises due to successive reproductive cell divisions.

Q16: A

NCERT gives examples like worker bees showing species survival does not depend on every individual reproducing.

Q17: B

Persistence of population across generations reflects species level reproduction.

Q18: C

Evolution depends on gene transmission which occurs only through reproduction.

Q19: A

Without accurate duplication of genetic material offspring cannot inherit parental traits.

Q20: D

Multicellular growth enlarges existing body but reproduction creates separate new individual.

Q21: A

Organisms that die soon after reproduction lack a post reproductive phase as per NCERT life cycle description.

Q22: B

NCERT states parrot has approximately 140 years lifespan whereas crow about 15 years elephant about 60 to 70 years and banyan tree several thousand years.

Q23: C

NCERT mentions fruit fly has a lifespan of approximately 20 to 30 days which is a few weeks.

Q24: C

Lifespan differs among species and environmental conditions can influence survival therefore R is false.

Q25: D

NCERT cites banyan tree as having lifespan extending to several thousand years.

Q26: A

NCERT indicates lifespan is a characteristic feature of species largely governed by genetic makeup though environment influences survival.

Q27: B

Lifespan spans from birth through juvenile reproductive and senescent phases until natural death.

Q28: B

NCERT mentions crow has approximately 15 years lifespan while butterfly about one to two weeks.

Q29: A

NCERT describes lifespan as a characteristic of species therefore individuals show comparable lifespan range.

Q30: D

Elephant is mentioned in NCERT with approximate lifespan of 60 to 70 years.

Want Full Chapter (120+ Questions)?

Download RankUp App ■